
CENG 102 – Algorithms & Programming II

ASSIGNMENT

Assignment Title: Mini Student Management System

General Description:

In this project, students are expected to develop a basic student information system using the C programming language. The system should be capable of storing the first name, last name, student number, and three course grades (midterm, assignment, final exam) for up to 100 students and allow various operations to be performed on this data.

Through the program, users should be able to add new student information, list all students and their details, search for a student based on student number, update or delete records, sort students by grade point average (GPA), and save the sorted data.

Each student record will store the following details:

- Full Name (maximum 50 characters)
- Student numbers (an integer between 0 and 100)
- Three Grades (Midterm, Assignment, Final Exam – each an integer between 0 and 100)
- GPA (calculated as the average of the three grades)

Required Functionalities:

The program must implement the following features:

The program should start with a menu and prompt the user to select an operation.

1. File Creation

- A data file (.txt or .dat) should be created to store student records.
- The initial file should be empty and contain default values (e.g., 0 for integers, empty string ("") for text fields).

2. Add New Student

- The user should be prompted to enter full name, student ID, and three grades.
- The GPA should be calculated automatically with the following weights: Midterm 40%, Assignment 10%, and Final Exam 50%.
- The entered data should be saved in the file at the appropriate location based on student number.
- If a student with the same ID already exists, the entry should be rejected with a warning message.

3. List all students

- The program should display the full name, student number, grades, and GPA of all students already saved in the file.

4. Search, Edit, and Delete Student Records

- The user should be able to search for a student-by-student number.
- If a match is found, all information regarding the student should be displayed.
- The user should then be able to edit the record and save the changes or delete the student entirely from the file.

5. Sort Students

- Students should be sorted in descending order by GPA and displayed as a full list.
- If desired, the user should be able to save the sorted list as a separate file.

6. Exit

- The program should run via a menu-based interface and continue to operate until the user chooses to exit.
- After completing an operation or encountering invalid input, the program should return to the previous (or any appropriate) menu.

Technical Requirements

- The program must include error handling to prevent crashes and instead inform the user, then return to the previous (or any appropriate) menu.
- Pointers, functions, structures, and file operations **must** be used meaningfully and effectively.
- Functions and variables name should be meaningful.
- The code should be explained briefly through comment lines where necessary (in English).
- The code must be compilable and executable; otherwise, it is going to be graded 0.

Submission Requirements:

- The project must be completed in groups of 2 or 3 students.
 - A list of groups is attached to this file. No changes will be made to the groups under any circumstances, regardless of the reason.
- The project should be submitted as a single .c file, and only one member of the group should submit it via email before the deadline.
- The top of the .c file must include the student numbers and full names of all group members in comment lines.
- **File Naming Format:** ceng102_group_groupNo_hw.c
 - *example:* ceng102_group_92_hw.c
- **Submission deadline:** 19 April 2026, Sunday – 23:59
 - *Late submissions will not be accepted or graded.*
- **Submission email:** ikchusnaibis@gmail.com
 - *Submissions to any email address other than the above provided will not be accepted or graded.*

Demo Presentation:

- Groups will present their project on 29 April 2025, Wednesday, for a brief 5 to 10 minutes demo. (Further details regarding the schedule and location will be announced later.)
- All members of the group are responsible for the entire project.
- The demo presentation will be conducted using the code directly. No additional presentation slides are required. Demo will be run on the code.
- The demo is mandatory, and all group members must attend. Projects without a demo will not be evaluated, and members who do not attend will be considered as not having submitted the assignment.
- Group members who cannot explain the project or provide satisfactory answers during the demo will receive significantly lower grades.

NOTE: Projects found to contain plagiarism or copied content will **not be evaluated under any circumstances**. Responsibility for such issues lies entirely with the students.

Grading:

10 pts – File Creation	10 pts – Sort Students
10 pts – Add New Student	5 pts – Menu System and Flow
10 pts – List Students	10 pts – Error Handling and User Feedback
10 pts – Search Students	10 pts – Effective use of Pointers, Functions, Structs, and File Operations
10 pts – Edit Students	5 pts – Code Quality (e.g. naming conventions and use of comment lines)
10 pts – Delete Students	And penalty scores in demo presentation...

Group	Student No	Name	Surname	Sec.
1	240401105	Yunus Emre	Uygun	2
	230401079	Halil	Taşyürek	2
2	250401016	Rozerin	Sun	1
	240401082	Esma Nur	Noyan	1
3	240401091	Nisa Nur	Sarı	1
	240401064	Sude Nur	Eger	1
4	240401072	Abdullah	Karaman	1
	240401101	Erkin	Tilkioğlu	1
5	240401078	Talha	Korkmaz	2
	250401104	Bilge	Zeybek	2
6	250401106	Hamza	Aşıkoğlu	1
	250401107	İsmail	Kalay	1
7	220401047	Hikmet Berke	Dindar	1
	250401019	Mehmet Boran	Yıldırım	1
8	240401050	Hasan Onur	Can	1
	240401087	Burak	Pala	1
9	250401015	Ege	Sonbahar	1
	250401012	Görkem	Sezen	1
	240401037	Efe	Arslan	1
10	230401096	İbrahim	Ayanoğlu	1
	220401041	İbrahim	Çelik	1
11	240401047	Ayberk	Bulut	1
	240401077	Yakup	Kıymaz	1
	240401081	Orkun	Mutaf	1
12	240401070	Ali	Güneyli	1
	240401076	Ali	Kılınç	1
13	250401014	Sena	Boncuk	1
	240401112	Yaren	Yıldırım	1
	240401079	Musa	Korkut	1
14	240401062	Halil İbrahim	Doymuş	1
	240401108	Görkem	Vuruşkan	1
	240401110	Bahadır	Yeniçay	1
15	240401053	Beyza	Çavaş	1
	240401045	Feyzanur	Bektaş	1
16	240401008	Muhammet Hüseyin	Coşkun	1
	240401052	Bertuğ Mehmet	Ceylan	2
17	240401059	Emre	Demir	1
	240401100	Uğur	Temel	1
18	240401068	Ege	Erpul	2
	240401129	Fatımanur	Güler	1
19	240401109	Zehra Erdem	Yalçınkaya	1
	240401106	Mert	Ünal	1
20	240401114	Enis Mert	Ziya	1
	240401040	Yiğit	Ayakdaş	1
21	240401034	Süleyman Ege	Aktaş	1
	230401085	Ahmet Buğra	Yanga	1
	230401022	Olcan Osman	Akyüz	1

Group	Student No	Name	Surname	Sec.
22	240401083	Enes	Oral	1
	240401084	Emirhan	Orhan	1
	240401092	Arda Ahmet	Sevin	1
23	250401030	Sıla Yüksel	Apaydın	1
	250401037	Ezgi Dilan	Çankaya	1
	250401073	Ahsen Sıla	Kuru	1
24	250401085	Eylül Tuana	Sulukaya	1
	250401099	Ceyda	Yıldız	1
	250401039	Nihal	Çelik	1
25	240401102	Halime Cemre	Tunçay	1
	240401075	Aybüke	Kılavuz	1
	240401043	Ceren	Bakışkan	1
26	240401080	Mustafa Emir	Küçük	1
	250401063	Hüseyin Efe	Kaya	1
27	250401083	Alperen	Sarı	1
	250401062	Kağan Can	Kaya	1
28	240401042	Hayriye Dilvin	Aydoğan	1
	250401069	Murat	Koç	1
29	240401086	Elanur	Özcan	1
	240401046	Can	Beran	1
30	240401089	Sinan	Peker	1
	240401036	Muhammet Emin	Aras	1
	240401061	Harun	Demirdağ	1
31	230401117	Uğurcan	Baçoğlu	1
	220401046	Barış	Dereioğlu	1
	220401053	Yalın	Gedik	1
32	250401052	Abidin Aras	Güler	2
	250401053	Kamil Kaan	Günay	2
	250401079	Berkay Eren	Onaç	2
33*	240401126	Semiha	Ünal	1
	240401051	Esma	Canbaşı	1
34*	250401006	Mahammad	Nazarlı	2
	250401013	Yiğit	Özel	1
	250401021	Gültekin	Dal	1
35*	240401049	Serdar	Büküm	1
	240401127	Berkant	Ünlü	1
	240401074	Ali	Kaymak	2
36*	250401089	Berke	Şenol	2
	250401092	İbrahim Alp	Turan	2
	250401095	Deniz Kağan	Uysal	2
37*	230401070	Tufan	Mudarasız	1
	230401089	Behzat	Yıldırım	1
	230401094	Halil Kerim	Yücel	1
38*	220401071	Aydın	Sarı	1
	220401090	Mehmet Buğra	Yurttaş	1
	230401041	Gökhan	Demirci	2
39*	210401013	İbrahim Erol	Çakıcı	1
	210401061	Samet Kağan	Demir	1

Groups marked with an asterisk (*) are formed randomly from students who did not declare their groups.